

1 / 20 | Ages 9-12 | 15 minutes

## Scale a colour recipe

### What we are practising

Scale both parts of a ratio by the same factor.

### What you need

Paper / Two coloured pencils or labelled symbols

Use B and Y labels instead of colours.

### Before you begin

Write a pretend recipe: 2 blue scoops and 3 yellow scoops make one batch.

This uses drawn scoops only; no paint mixing or tasting is needed.

### Try it together

1. Draw the five scoops in one batch.
2. Find how many complete batches make 20 scoops.
3. Calculate the blue and yellow amounts for that total.
4. Check that the parts sum to 20 and keep the original ratio.

### Talk, draw or show your idea

Would adding the same number to both parts preserve the recipe?

### Space to explore

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2 / 20 | Ages 9-12 | 20 minutes

## Chance versus results

### What we are practising

Distinguish theoretical probability from observed frequency.

### What you need

Six equal paper cards / Opaque bag or bowl with cloth cover / Pencil

An adult shuffles the cards face down and offers a blind selection.

### Before you begin

Label three cards A, two B and one C; fold them identically.

Use large paper cards and keep bags away from faces and younger children.

### Try it together

1. Predict the chance of drawing each letter.
2. Mix, draw one card without looking, record it and put it back.
3. Repeat for 18 draws, remixing every time.
4. Compare counts with the predicted proportions.

### Talk, draw or show your idea

Must A appear exactly nine times in eighteen draws?

### Space to explore

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3 / 20 | Ages 9-12 | 15 minutes

## Which notebook offer?

### What we are practising

Calculate unit prices and distinguish cheapest per item from affordable today.

### What you need

Paper / Pencil

Draw groups of notebooks and distribute pretend price tokens.

### Before you begin

Write two offers: 3 notebooks for 12 tokens; 5 notebooks for 18 tokens. Budget: 15 tokens.

Use fictional tokens; this activity requires no purchase or payment details.

### Try it together

1. Calculate the price per notebook for each bundle.
2. Identify which bundle has the lower unit price.
3. Decide which complete bundle fits the budget; bundles cannot be split.
4. Explain why the two answers differ.

### Talk, draw or show your idea

Can the better unit price still be unaffordable?

### Space to explore

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4 / 20 | Ages 9-12 | 20 minutes

## Same area, different fence

### What we are practising

Distinguish area from perimeter using factor pairs.

### What you need

Squared paper or a drawn grid / Pencil

Draw labelled rectangles without exact grid squares.

### Before you begin

Imagine a garden of 24 square units with whole-number side lengths.

Use paper plans only; no fencing, digging or outdoor construction is involved.

### Try it together

1. List the factor pairs of 24 to find possible rectangles.
2. Draw or label each rectangle and confirm its area.
3. Calculate each perimeter by adding all four sides.
4. Choose the rectangle needing the least fence among these options.

### Talk, draw or show your idea

Why can equal areas have unequal perimeters?

### Space to explore

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5 / 20 | Ages 9-12 | 15 minutes

## Which average tells the story?

### What we are practising

Calculate two summaries and explain the effect of an unusually large value.

### What you need

Paper / Pencil

Use drawn stacks of squares to represent the values.

### Before you begin

Write five fictional reading-session lengths in minutes: 2, 3, 3, 4, 13.

Use fictional numbers; do not compare children's ability or worth using these measures.

### Try it together

1. Add the five values and divide by five to find the mean.
2. Find the middle value in the ordered list to get the median.
3. Replace 13 with 3 and calculate both again.
4. Explain which summary changed more and why.

### Talk, draw or show your idea

Does an average have to equal one of the actual measurements?

### Space to explore

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6 / 20 | Ages 9-12 | 10 minutes

## A quarter as a percentage

### What we are practising

Calculate twenty-five percent by finding one quarter.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A box contains 80 cards. Exactly 25% are blue. How many blue cards are there?

### Space to explore

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7 / 20 | Ages 9-12 | 10 minutes

## Add the decimal lengths

### What we are practising

Align decimal place values when adding lengths.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

Two ribbons are 2.75 metres and 1.6 metres long. What is their total length, with no overlap or cutting?

### Space to explore

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8 / 20 | Ages 9-12 | 10 minutes

## Combine thirds and sixths

### What we are practising

Add fractions by converting to a shared denominator.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A jug has two thirds of a litre of water. One sixth of a litre is added, with none spilled. How much water is there?

### Space to explore

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9 / 20 | Ages 9-12 | 10 minutes

## A triangle from a rectangle

### What we are practising

Find a triangle's area as half a matching rectangle.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A right triangle is made by cutting an 8 cm by 5 cm rectangle along a diagonal into two equal-area triangles. What is one triangle's area?

### Space to explore

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10 / 20 | Ages 9-12 | 10 minutes

## Count the cubes in the layers

### What we are practising

Calculate a rectangular prism's volume by layers.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A solid model has four identical layers. Each layer is a rectangle of three cubes by two cubes, with no gaps. How many unit cubes form the model?

### Space to explore

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11 / 20 | Ages 9-12 | 10 minutes

## The chance of not choosing yellow

### What we are practising

Find a complementary probability in an equally likely draw.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A bag contains seven identical-feeling counters: two yellow and five green. One is drawn without looking, each equally likely. What is the probability it is not yellow?

### Space to explore

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12 / 20 | Ages 9-12 | 10 minutes

# Find the spread of the scores

## What we are practising

Calculate range from the largest and smallest data values.

## What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

## Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

## Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

## Talk, draw or show your idea

Five practice scores are 4, 7, 7, 10 and 12. The range is the largest minus the smallest. What is the range?

## Space to explore

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13 / 20 | Ages 9-12 | 10 minutes

## Two fictional city clocks

### What we are practising

Add an hour-and-minute offset to a clock reading.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

In a fictional puzzle, City B is exactly three hours and thirty minutes ahead of City A. It is 10:45 in A. What is the 24-hour time in B?

### Space to explore

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14 / 20 | Ages 9-12 | 10 minutes

## Undo the number machine

### What we are practising

Solve a two-step equation by inverse operations.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A machine triples a whole number, then adds five. Its output is 26. What was the input?

### Space to explore

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15 / 20 | Ages 9-12 | 10 minutes

## From below zero to above

### What we are practising

Find a change across zero using a number line.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

In a maths story, a thermometer rises from  $-3^{\circ}\text{C}$  to  $5^{\circ}\text{C}$ . By how many degrees does it rise?

### Space to explore

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16 / 20 | Ages 9-12 | 10 minutes

## Distance along one grid row

### What we are practising

Find horizontal distance from coordinates with equal heights.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

On a square grid, A is at (2, 4) and B at (7, 4). The first coordinate is horizontal. How many grid units separate A and B along that row?

### Space to explore

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17 / 20 | Ages 9-12 | 10 minutes

## Distance at a steady speed

### What we are practising

Multiply a constant rate by elapsed time.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

In a pretend route, a cyclist travels steadily at 12 kilometres per hour for one and a half hours, with no stops. How far does the cyclist travel?

### Space to explore

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18 / 20 | Ages 9-12 | 10 minutes

## Mass remaining in grams

### What we are practising

Convert kilograms to grams before subtracting.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A bag in a maths problem contains 1.2 kilograms of rice. After 350 grams are used, how many grams remain? One kilogram is 1,000 grams.

### Space to explore

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19 / 20 | Ages 9-12 | 10 minutes

## Two names for the same fraction

### What we are practising

Recognise equivalent fractions by scaling numerator and denominator.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

Which of  $\frac{6}{8}$ ,  $\frac{9}{12}$  and  $\frac{3}{8}$  are equal to  $\frac{3}{4}$ ? More than one choice may fit.

### Space to explore

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20 / 20 | Ages 9-12 | 10 minutes

## The missing angle in a turn

### What we are practising

Use a full turn of 360 degrees to find a missing angle.

### What you need

Paper / Pencil

Explain aloud or ask an adult to record your ideas.

### Before you begin

Read the question first; use paper for a sketch or calculation.

An adult is available for unfamiliar vocabulary or checking a calculation; no personal information is needed.

### Try it together

1. Read the question below and identify what you need to find.
2. Try an answer; show your thinking with words, fingers or a quick drawing.
3. Use the hint if needed, then compare with the answer and explain why it fits.

### Talk, draw or show your idea

A robot turns  $90^\circ$  clockwise, then  $120^\circ$  clockwise. How many more degrees clockwise complete exactly one full  $360^\circ$  turn?

### Space to explore

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# For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

## 1. Scale a colour recipe

### A little hint

Twenty scoops is four batches of five.

### Answer / what to notice

Use 8 blue scoops and 12 yellow scoops; both original amounts are multiplied by 4.

Equal multiplication preserves the ratio; adding the same amount generally does not.

### Make it easier

Scale the recipe to 10 scoops.

### Take it further

Find the amounts for 7.5 scoops using fractional batches.

### Another day, away from the screen

Invent a bead-pattern ratio on paper and scale it to two different totals.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 2. Chance versus results

#### A little hint

Each draw starts with all six cards again.

#### Answer / what to notice

A has chance  $\frac{3}{6}$ , B  $\frac{2}{6}$  and C  $\frac{1}{6}$ . Counts near 9, 6 and 3 are plausible, but exact counts are not guaranteed.

Replacing and mixing restores the same chances; random short runs can differ from expected proportions.

#### Make it easier

Compare just two A cards and two B cards over eight draws.

#### Take it further

Combine results from several rounds and compare relative frequencies.

#### Another day, away from the screen

Design a different card mix and predict which outcomes become more likely.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 3. Which notebook offer?

#### A little hint

Divide the total price by the number of notebooks, then compare the full price with 15.

#### Answer / what to notice

The unit prices are 4 and 3.6 tokens. The five-pack is cheaper per notebook, but only the three-pack fits 15 tokens.

Unit price compares value for equal quantities; the total payment determines whether the budget covers the purchase.

#### Make it easier

Use 2 notebooks for 8 and 4 for 12.

#### Take it further

Find the lowest total cost to obtain at least eight notebooks using these whole bundles.

#### Another day, away from the screen

Invent two offers where the larger bundle has a worse unit price.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 4. Same area, different fence

#### A little hint

The distinct pairs are  $1 \times 24$ ,  $2 \times 12$ ,  $3 \times 8$  and  $4 \times 6$ .

#### Answer / what to notice

Their perimeters are 50, 28, 22 and 20 units; the 4 by 6 rectangle needs the least fence.

Area counts the inside squares; perimeter measures the outside boundary, which changes with shape.

#### Make it easier

Compare only 3 by 8 and 4 by 6.

#### Take it further

Repeat for area 36 and compare the most compact whole-number rectangle.

#### Another day, away from the screen

Look for two arrangements of twelve paper squares with different perimeters.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 5. Which average tells the story?

#### A little hint

The first total is 25; the middle of five ordered values is the third.

#### Answer / what to notice

Original mean 5, median 3. Revised values 2, 3, 3, 3, 4 have mean 3 and median 3.

The mean uses every value's size; the median depends on the ordered middle position, so one large value affects them differently.

#### Make it easier

Find only the median using five number cards.

#### Take it further

Create a five-number dataset with median 4 and mean 6.

#### Another day, away from the screen

Compare mean and median for invented game scores and explain what each reveals.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 6. A quarter as a percentage

#### A little hint

25% means one quarter of the total.

#### Answer / what to notice

20 blue cards.

$80 \div 4 = 20$ ; equivalently,  $80 \times 25 \div 100 = 20$ .

#### Make it easier

Split eighty into four equal groups.

#### Take it further

How many cards would be blue if 75% were blue?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 7. Add the decimal lengths

#### A little hint

Write 1.6 as 1.60 so the decimal places align.

#### Answer / what to notice

4.35 metres.

$2.75 + 1.60 = 4.35$ ; hundredths are added to hundredths.

#### Make it easier

Add whole metres first, then the remaining hundredths.

#### Take it further

How much more ribbon would make a total of five metres?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 8. Combine thirds and sixths

#### A little hint

Two thirds equals four sixths.

#### Answer / what to notice

Five sixths of a litre.

$\frac{2}{3} = \frac{4}{6}$ , and  $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$ . The denominator stays six because the parts are sixths.

#### Make it easier

Draw a litre strip divided into six equal sections.

#### Take it further

How much more would fill a one-litre jug?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 9. A triangle from a rectangle

#### A little hint

Find the rectangle's area, then halve it.

#### Answer / what to notice

20 square centimetres.

The rectangle has area  $8 \times 5 = 40 \text{ cm}^2$ . Each of its two equal triangles has  $40 \div 2 = 20 \text{ cm}^2$ .

#### Make it easier

Draw the diagonal and label the two equal halves.

#### Take it further

What happens to the triangle's area if the rectangle's length doubles?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 10. Count the cubes in the layers

#### A little hint

Find the cubes in one layer before multiplying by four.

#### Answer / what to notice

24 unit cubes.

Each layer has  $3 \times 2 = 6$  cubes, and four layers have  $6 \times 4 = 24$ .

#### Make it easier

Sketch the six cubes of one layer.

#### Take it further

How many cubes are added by one more identical layer?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 11. The chance of not choosing yellow

#### A little hint

Not yellow includes all five green counters.

#### Answer / what to notice

Five sevenths, or  $5/7$ .

Five of seven possible counters meet the condition. This is also  $1 - 2/7 = 5/7$ .

#### Make it easier

Draw seven circles and mark the two yellow ones.

#### Take it further

If one yellow counter is removed before drawing, what is the new probability of green?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 12. Find the spread of the scores

#### A little hint

Use only the endpoints, not the total.

#### Answer / what to notice

Eight points.

$12 - 4 = 8$ . The repeated seven does not change either endpoint.

#### Make it easier

Circle the smallest and largest values first.

#### Take it further

If a new score of fifteen is added, what is the new range?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 13. Two fictional city clocks

#### A little hint

Add three hours, then thirty minutes.

#### Answer / what to notice

14:15.

10:45 plus three hours is 13:45; thirty more minutes gives 14:15.

#### Make it easier

Use a drawn clock or number line for the minutes.

#### Take it further

If B reads 09:00, what does A read under the same rule?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 14. Undo the number machine

#### A little hint

Undo the final addition before undoing the tripling.

#### Answer / what to notice

Seven.

$26 - 5 = 21$ ;  $21 \div 3 = 7$ . Checking gives  $3 \times 7 + 5 = 26$ .

#### Make it easier

Write the reverse path: subtract five, divide by three.

#### Take it further

What input would produce an output of thirty-two?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 15. From below zero to above

#### A little hint

Count the rise to zero, then from zero to five.

#### Answer / what to notice

Eight degrees.

Three degrees reach zero and five more reach five:  $3 + 5 = 8$ .

#### Make it easier

Mark every integer from minus three to five.

#### Take it further

What is the change when it falls from five to minus one?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 16. Distance along one grid row

#### A little hint

The second coordinate matches, so compare the first coordinates.

#### Answer / what to notice

Five grid units.

Both points lie on the row with second coordinate four;  $7 - 2 = 5$  horizontal units.

#### Make it easier

Draw the row and label positions two through seven.

#### Take it further

How far apart are (2, 4) and (2, 9) vertically?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 17. Distance at a steady speed

#### A little hint

Find the distance for one hour and for half an hour.

#### Answer / what to notice

18 kilometres.

One hour gives 12 km and half an hour gives 6 km;  $12 + 6 = 18$ .

#### Make it easier

Split the time into three half-hour blocks.

#### Take it further

At the same constant speed, how long would 30 kilometres take?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 18. Mass remaining in grams

#### A little hint

Convert 1.2 kilograms to 1,200 grams first.

#### Answer / what to notice

850 grams.

$1.2 \times 1,000 = 1,200$ ; then  $1,200 - 350 = 850$  grams.

#### Make it easier

Subtract three hundred, then fifty.

#### Take it further

How many more grams must be used to leave half a kilogram?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 19. Two names for the same fraction

#### A little hint

Multiply both the top and bottom of  $\frac{3}{4}$  by the same number.

#### Answer / what to notice

$\frac{6}{8}$  and  $\frac{9}{12}$ .

Multiplying both parts by two gives  $\frac{6}{8}$ ; by three gives  $\frac{9}{12}$ . Changing only the denominator to eight gives a different value.

#### Make it easier

Test  $\frac{6}{8}$  first by splitting each quarter in two.

#### Take it further

Give another equivalent fraction and explain the factor used.

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 20. The missing angle in a turn

#### A little hint

Add the completed turns, then subtract from 360.

#### Answer / what to notice

150 degrees.

$90 + 120 = 210$ ;  $360 - 210 = 150$ . All turns use the same direction.

#### Make it easier

Draw a circle and mark the completed total of 210 degrees.

#### Take it further

Would reversing direction on the second turn give the same remaining clockwise angle?

#### Another day, away from the screen

Another day, change one detail and explain whether the answer changes.